

PREPROCESSING REPORT ON KITWE NEWS PROJECT

Introduction

The preprocessing stage for the Kitwe-Focused News Project follows the data collection phase, during which news articles centered on Kitwe, Zambia were gathered from various Zambian news sources via RSS feeds. The objective of this stage is to cleanse, standardize, and label the collected data, transforming it into a format suitable for further analysis, including the classification of news authenticity. The report details the methodologies, processes, and tools employed to establish a robust preprocessing pipeline, ensuring that the dataset is accurate, standardized, and easily accessible for the next phases of the project.

Preprocessing Objectives

The primary goals of the preprocessing pipeline include:

1. **Data Cleansing:** Remove irrelevant and incorrect information to ensure accuracy and reliability.
2. **Text Preprocessing:** Standardize and prepare text data, facilitating machine learning and keyword extraction.
3. **Categorization and Labeling:** Classify each news entry as real or fake based on predefined criteria.
4. **Task automation:** Automate the preprocessing task through a data pipeline to ensure consistency and streamline repetitive tasks.

Preprocessing Pipeline Approach

The preprocessing stage is designed to modularize tasks, with each step encapsulated in standalone Python modules. This structure promotes maintainability, testing, and reuse. The main script orchestrates module execution, allowing a seamless data flow through the pipeline. Team members can either consolidate existing work into modules or contribute by developing new components where needed.

Pipeline Components and Key Steps

1. **Data Cleansing**
 - **Objective:** Remove inaccuracies, duplicates, and irrelevant data.
 - **Implementation:**
 - Identify and remove incomplete or inconsistent entries.
 - Filter out duplicate articles, focusing on unique news entries to maintain data integrity.
 - **Result:** A cleaner dataset free from redundant or incorrect information, enhancing overall data quality.
2. **Text Preprocessing**
 - **Objective:** Standardize and clean text data for analysis and classification.
 - **Implementation:**
 - Tokenization, lowercasing, punctuation removal, and stopword elimination.
 - Handle symbols that could misrepresent text content.

- **Result:** Text data is normalized and simplified, making it ready for analysis and LLM building.
- 3. **Categorization and Labeling**
 - **Objective:** Classify news articles as real or fake based on rule-based criteria.
 - **Implementation:**
 - **Source Credibility:** Check each article's source against a list of known, reputable outlets.
 - **Clickbait Detection:** Flag articles with excessive punctuation, provocative language, or full capitalization.
 - **Keyword Analysis:** Identify articles with a high frequency of sensational keywords.
 - **LDA Topic Modeling:** Use topic modeling to detect dubious topics, tagging articles as potentially sensational.
 - **Sentiment Analysis:** Flag articles with extreme sentiment scores as potential indicators of fake news.
 - **Other Enhanced Criteria:** Apply additional rules like excessive capitalization, vague authorship, and mismatched headline-description content.
 - **Result:** Each article receives a "Target_final" label (0 for fake, 1 for real), contributing to a refined dataset for subsequent analysis.
- 4. **Task automation**
 - **Objective:** Package and automate pipeline tasks for consistency and repeatability.
 - **Implementation:**
 - **Airflow:** Schedule and orchestrate tasks, ensuring regular data updates..
 - **Result:** A reproducible and automated pipeline that minimizes manual intervention and ensures consistency in data preprocessing.

Challenges and Solutions

Verifying Fake News Sources

Solution: A source credibility check was implemented using a predefined list of reputable and suspicious sources. A custom function (updated_check_source_credibility) categorized each source as reputable, suspicious, or unknown. This helped us filter out potentially unreliable sources, strengthening the overall credibility of our dataset.

Conclusion

The data preprocessing phase for the Kitwe-Focused News Project successfully transformed raw data into a structured, labeled, and accessible format. By consolidating and automating tasks through a modular pipeline, we achieved an efficient and reliable data foundation.

The link to the code is below :

<https://github.com/EdBali/KitweNewsProject>